

SNIA SFF spec updates to the Storage Form Factor

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It’s Been a Busy Year…

- Lots of questions that we needed to answer
  - How is an SSD liquid cooled?
  - How does an SSD support Streaming Boot?
  - How does and SSD support PCIe 7.0 throughput?
  - Why are there I3C problems?
  - How is a liquid cooled SSD tested?

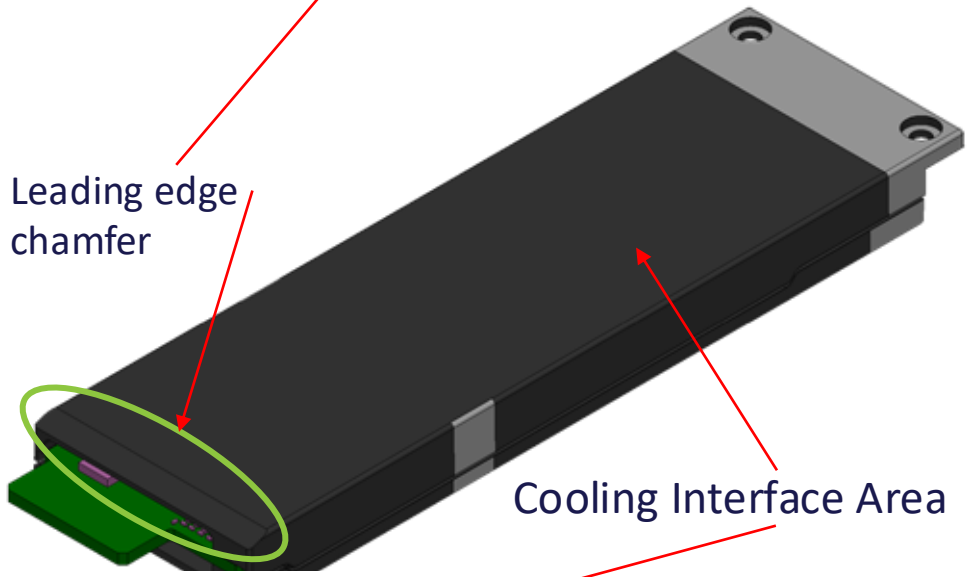
- My goal today is to provide these answers
  - Or at least tell you what progress has been made

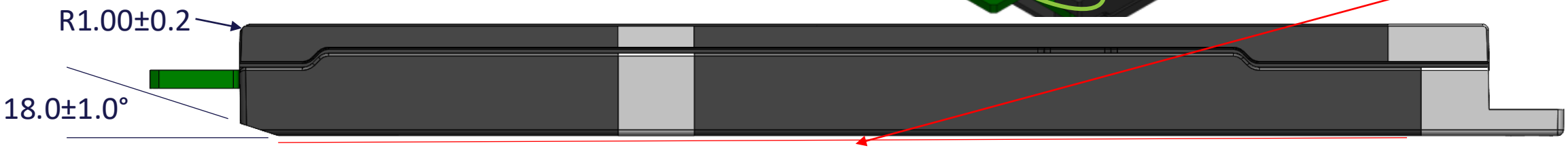
Liquid cooled E1.S (SFF-TA-1006)

Rev 2.0: https://members.snia.org/document/dl/26956

1. Defined "Cooling Interface Area" on secondary side for 9.5mm
   - Supports air-cooled and direct liquid cooling
   - Mating surface between device and cold plate.
2. Add leading edge chamfer on secondary side and leading-edge radius on primary side
   - Aids with insertion
3. Flatness and roughness specs on Secondary side
   - Flatness: < 0.12 mm
   - Roughness: <1.6 um
   - Improves cooling efficiency between device and cold plate
4. Added max pressure (138 kPA) supported on Cooling Interface Area
   - Prevents damage when force applied



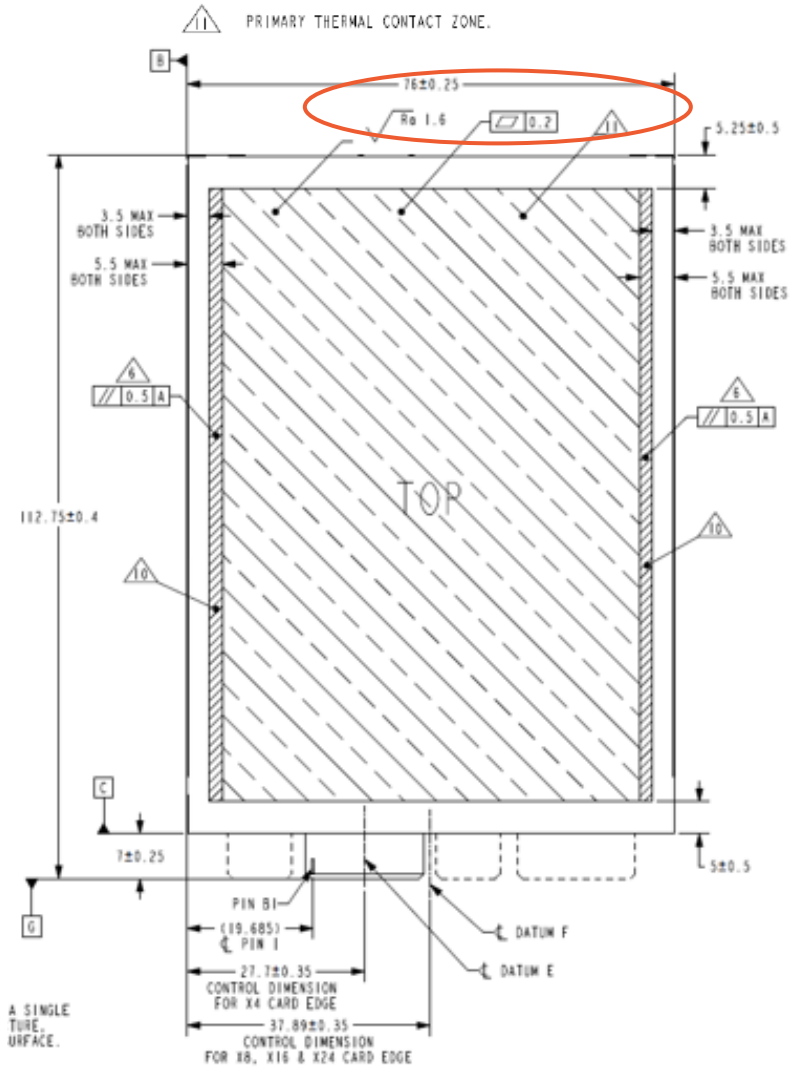




Liquid cooled E3 (SFF-TA-1008)

Rev 3.0: https://members.snia.org/document/dl/26716

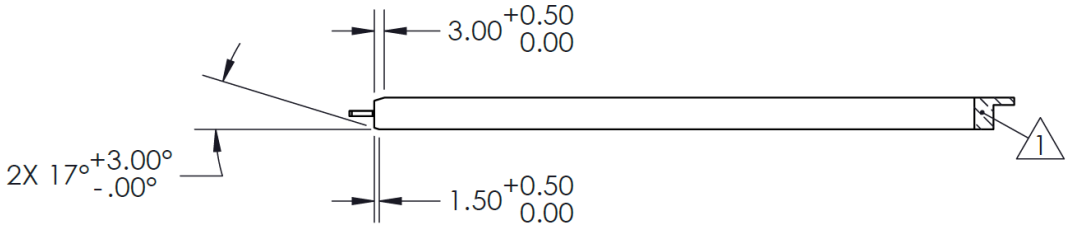
1. Defined "Cooling Interface Area" on top side for 7.5mm E3.S and E3.L
   - Supports air-cooled and direct liquid cooling
   - Mating surface between device and cold plate.
2. Label on **bottom** side only
   - Improves cooling efficiency of SSD and label durability
3. Flatness and roughness specs on top side
   - Flatness: < 0.2 mm (E3.S), < 0.25 mm (E3.L)
   - Roughness: <1.6 um
   - Improves cooling efficiency between device and cold plate
4. Added max pressure (103 kPA) supported on Cooling Interface Area
   - Prevents damage when force applied

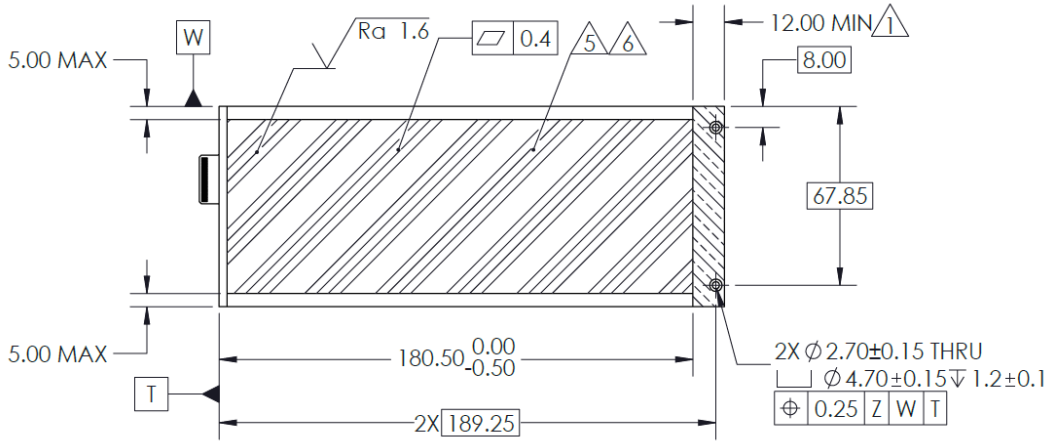


Liquid cooled E2 (SFF-TA-1042)

- In development:
  - Details at https://members.snia.org/document/dl/59144

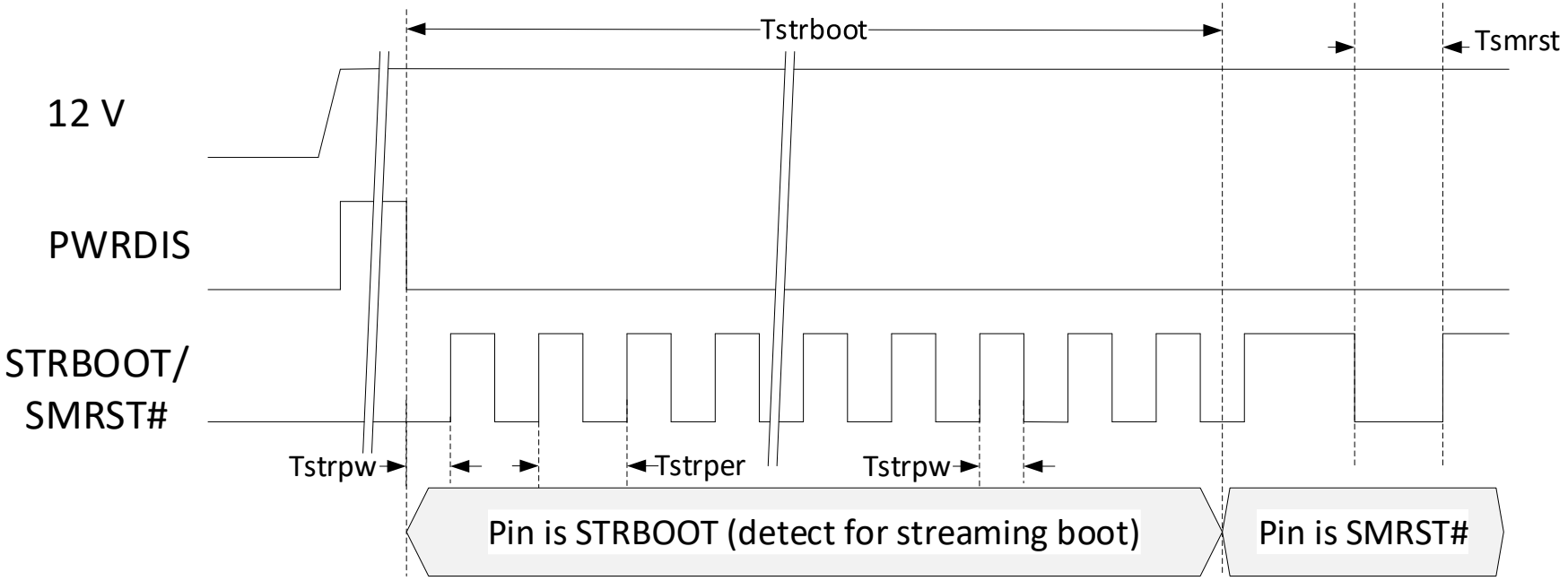
1. Defined "Cooling Interface Area" on secondary side
   - Supports air-cooled and direct liquid cooling
   - Mating surface between device and cold plate.
2. Add leading edge chamfer on both sides
   - Aids with insertion
3. Flatness and roughness specs on Secondary sid
   - Flatness: < 0.40 mm
   - Roughness: <1.6 um
   - Improves cooling efficiency between device and cold plate
4. Added max pressure (69 kPA) supported on Cool Interface Area
   - Prevents damage when force applied





EDSFF Pin/Signal Spec (SFF-TA-1009): Streaming Boot

- Intent is to enable Streaming Boot mode (Boot from Host)
- To do this, we overloaded the SMRST# pin with this function.
  - If the host wants Streaming boot mode, it will send a pattern within the first 100ms of power good to the device.



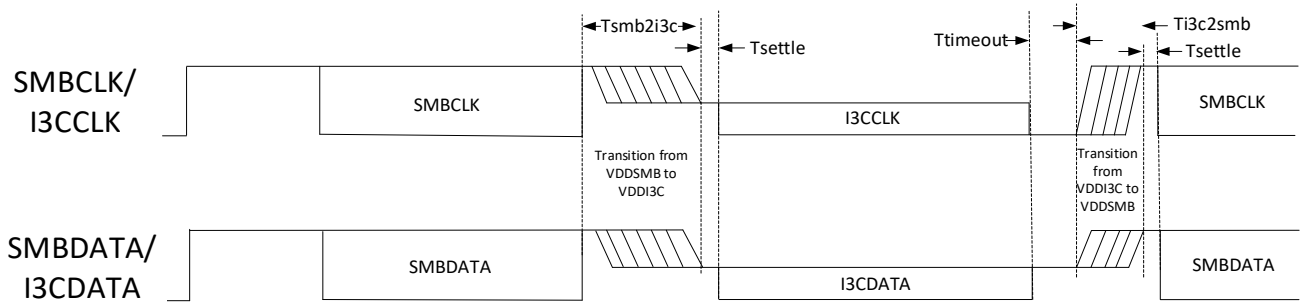
EDSFF Pin/Signal Spec (SFF-TA-1009): PCIe 7.0

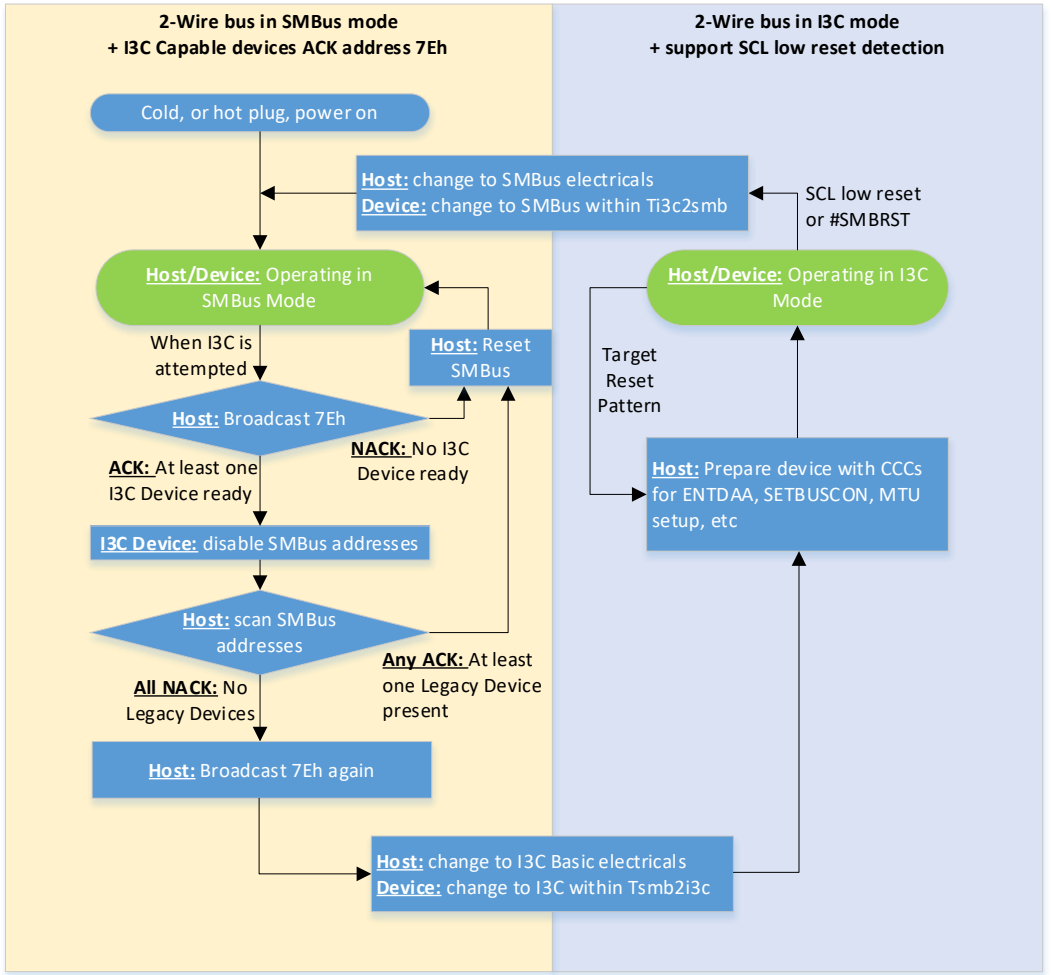
- Intent is to support PCIe 7.0 data rates
- At Nyquist:
  - IL = -7 dB
  - RL = -7.33 dB
  - PSNEXT = -60 dB
  - PSFEXT = -50 dB

Line Rate	Insertion Loss (IL)	Return Loss (RL)	Power Sum Near End Crosstalk (PSNEXT)[1]	Power Sum Far End Crosstalk (PSFEXT)[1]
128.0 GT/s	≥ -2 - 0.15625 * f dB (f = 0 to 32 GHz) ≥ 1 - 0.25 * f dB (f = 32 to 48 GHz)	≤ -10 + 0.0833 * f dB (f = 0 to 48 GHz)	≤ -60 dB (0 to 32 GHz) ≤ -70 + 0.3125 * f dB (f = 32 to 48 GHz)	≤ -50 + 0.104 * f dB (f = 0 to 48 GHz)

EDSFF Pin/Signal Spec (SFF-TA-1009): I3C

- Intent is to simplify and eliminate ambiguity
  - Changes to timing diagram
  - Changes to reference flow
  - Changes to informative wording





EDSFF Pin/Signal Spec (SFF-TA-1009): Timeline

- Spec revision is current 4.1.5
  - Found here https://members.snia.org/document/dl/26728
- Ideally we publish this specification by end of year (Revision 5.0)

Thermal Characterization Specification (SFF-TA-1023)

- Work started!
- What is planned for this specification:
  - New sections to clarify Air vs. DLC characterization
  - Thermal reporting of a Direct Liquid Cooled device.
  - Thermal data collection and testing for a Direct Liquid Cooled device.
  - Reference test fixtures

Call to Action

- Let’s have some open conversations in the OCP Storage project
  - Liquid Cooling PCIe 7.0?
  - SI topologies for PCIe 7.0 (cabling)?

- Continue driving the form factor evolution
  - Complete the draft specs
  - Add USB 2.0 to EDSFF

- Where to find additional information
  - OCP Storage: http://www.opencompute.org/wiki/Storage
  - EDSFF Specifications: https://www.snia.org/sff/specifications

Thank You!

